

EXPERIMENT - 13

greatest of 2 numbers.

AIM:

To write an assembly language program to find the smallest number in an array using 8085 microprocessor in 8085.

software used:

8085 simulator.

ALGORITHM:

- 1) Initialize the count.
- 2) Get the input numbers.
- 3) compare the content of accumulator (A) with HL pair for all input numbers.
- 4) Store smallest number in output register.
- 5) End program.

PROGRAM:

MNEMONICS	EXPLANATION
LDA 2050	Load content of memory 2050H into A.
MOV B, A	Move content of A into B.
LDA 2051	Load content of memory 2051 into A.

CMP B

compare content of register B with A.

JNC STORE.

Jump to store if no carry, which means $A \geq B$.

MOV A, B

If there is a carry. (i.e., $A < B$), move B into A.

STORE: STA 2052

store content of A into 2052.

HLT

halt.

INPUT:

Address	Data
2050	29
2051	22

OUTPUT:

Address	Data
2052	29

RESULT:

This is assembly language program to find smallest number using 8085 simulator.



Registers			Flag	
A	00		S	0
BC	00	00	Z	1
DE	00	00	AC	0
HL	00	00	P	1
PSW	00	00	C	0
PC	42	10		
SP	FF	FF		
Int-Reg	00			

Load me at

```
1 LDA 2050H
2 MOV B, A
3 LDA 2051H
4 CMP B
5 JNC STORE
6 MOV A, B
7 STORE: STA 2052H
8 HLT
```

Decimal - Hex Conversion

Decimal	Hex
0	0
→ To Hex	← To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

2051	-	+	16
Update Memory			

Data Stack KeyPad **Memory** I/O Ports

Start 2052

OK

Address (Hex) Address Data

0804	2052	29
0805	2053	0
0806	2054	0
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0
0810	2064	0
0811	2065	0
0812	2066	0
0813	2067	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

